

KAPIL

AI Engineer

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PROFESSIONAL SUMMARY

Software Engineer (2.7+ years) transitioning into AI Engineering, with hands-on focus on LLM integration, Retrieval-Augmented Generation, and agentic systems built with LangChain and LangGraph. Studying and applying core ML/DL foundations, vector and graph databases (Qdrant, Neo4j), and production system design for LLM applications — observability, cost optimization, and scalable Python backends. Currently completing an M.Tech thesis on Multimodal Sentiment Analysis using Deep Learning (CMU-MOSI/MOSEI), alongside a portfolio of production-style AI systems: a hybrid-retrieval assistant (Second Brain) and a checkpointed multi-agent research pipeline. Brings a strong engineering foundation — real-time systems, secure APIs, scalable backend architecture — to AI system design.

TECHNICAL SKILLS

AI & Machine Learning: LLM Integration (OpenAI, Claude, Gemini), Prompt Engineering (zero/few-shot, chain-of-thought), Fine-Tuning vs RAG vs Prompting, Deep Learning (Transformers, Attention, Tokenization), Vector Embeddings & Semantic Search

RAG & Agents: RAG Pipelines (chunking strategies, hybrid dense+sparse search, re-ranking, RAGAS evaluation), LangChain, LangGraph, AI Agents (ReAct pattern, multi-agent orchestration, agent memory & checkpointing)

Vector & Graph Databases: Qdrant (HNSW indexing, quantization, hybrid search, payload filtering), Neo4j (Cypher, GraphRAG, knowledge graphs), MongoDB (Atlas Vector Search, LangGraph checkpointing)

Python & Systems: Python (asyncio, threading, multiprocessing, GIL), Concurrency (task queues, Celery, Redis pub/sub), Caching & Rate Limiting

Full-Stack Development: React.js, Node.js, Express.js, FastAPI, TypeScript, RESTful API Design, WebSockets

Databases: MongoDB, PostgreSQL, Qdrant, Redis

Production & MLOps: Docker, CI/CD, LLM Observability (LangSmith), Model Serving & Cost Optimization

PROJECTS

M.Tech Thesis — Multimodal Sentiment Analysis using Deep Learning (SLMFN)

Designing and evaluating a Small Language Model Fusion Network for multimodal sentiment classification, benchmarked on the CMU-MOSI and CMU-MOSEI datasets. Vaish College of Engineering, Rohtak (affiliated with MDU Rohtak).

Second Brain — Personal AI Assistant (In Progress)

Building a full-stack AI assistant with hybrid retrieval (Qdrant dense vectors + BM25 sparse search), a LangGraph tool-using agent (web search, document retrieval, calendar) with MongoDB-backed checkpointing for persistent conversation threads, and a FastAPI backend streaming SSE responses to a Next.js chat interface with document upload.

Multi-Agent Research Assistant — LangGraph Orchestration with Checkpointing

Architected a multi-agent research system (Planner, Researcher, Writer, Critic) with LangGraph, using MongoDB checkpointing to resume long-running research tasks, Qdrant for past research notes, and live web search (Tavily) to produce structured markdown reports.

PROFESSIONAL EXPERIENCE

Easyrewardz Software Services | Frontend Developer

Feb 2025 – Present

Gurugram, Haryana, India

- Web Push & Pop Notification System: Architected and implemented the entire system from scratch, including folder structure, file organization, and scalable Node.js modules to handle push and pop notifications.
- Secured RESTful API endpoints with signature-based authentication and input validation using Joi, reducing invalid requests by 95%. Integrated MySQL and MongoDB for persistent storage, and configured Kafka & Redpanda for high-throughput, fault-tolerant message queuing.
- Developed webhook handlers to capture and process real-time user interactions, improving engagement tracking, and built Socket.io modules enabling real-time bi-directional communication.
- StorePay SDK (Vanilla JavaScript & React) — Architected, developed, and published secure payment SDKs on npm, implementing an end-to-end encrypted payment flow within an embedded iFrame while maintaining PCI-compliant communication between merchant applications and the payment gateway.
- ZenceBotBuilder: Bot Builder Designer — Led full-stack development of a bot flow designer, defining GraphQL-like JSON schemas to represent live bot flows and building server-side APIs with Express, Prisma ORM, Redis, and Azure Blob Storage.

Tata Consultancy Services | Graduate Trainee → Programmer

Nov 2023 – Feb 2025

Trivandrum, Kerala → Gurugram, Haryana, India

- Completed structured training across coding, networking, and systems fundamentals before transitioning into a Programmer role delivering client-facing features.
- Delivered solutions across diverse client projects by translating business requirements into reliable, scalable code, collaborating with engineering, QA, and project stakeholders to remove blockers and align on outcomes.
- Leveraged AI-assisted tools and intelligent automation to boost productivity and maintain higher code quality while accelerating delivery cycles.
- Replaced manual testing of VCR conference room systems with an automated, AI-assisted scripting process, significantly reducing testing time and operational burden while improving service reliability.

EDUCATION

Master of Technology (M.Tech) — Computer Science

Aug 2024 – Jul 2026

Vaish College of Engineering, Rohtak (affiliated with MDU Rohtak) — Thesis: Multimodal Sentiment Analysis using Deep Learning